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MILITARY SPECIFICATION

TRANSMITTER, TEMPERATURE, ELECTRICAL RESISTANCE

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers the requirements for flexible sheath, electrical resistance type cylinder head temperature transmitters.

1.2 Classification. The transmitters shall be classified as follows:

MS24482-1 - 2.850 inch stem length

MS24482-2 - 2.203 inch stem length

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-A-225/5E	Aluminum Alloy 2017, Bar, Rod and Wire, Rolled Drawn, or Cold Finished.
QQ-S-781	Strapping, Steel, and Seals.
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Box, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated Plywood.
PPP-B-621	Box, Wood, Nailed and Lockcorner.
PPP-B-636	Box, Shipping, Fiberboard.
PPP-B-676	Boxes, Setup.

MILITARY

MIL-P-116	Preservation, Methods of.
DOD-D-1000	Drawing, Engineering and Associated List.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio Air Logistics Center, Kelly AFB TX 78241-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6685

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-143	Standards and Specifications Order of Precedence for the Selection of.
MIL-STD-147	Palletized Unit Loads.
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of.
MIL-STD-810	Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	Dissimilar Metals.
MS24482	Transmitter, Temperature, Electrical Resistance, Armored, Bayonet Locking Type (-50°C to 300°C).

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

AIR FORCE-NAVY AERONAUTICAL

AN4076	Fitting & Engine Cylinder Head Temperature measuring Element.

* (Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-government documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951	Packaging, Commercial.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL BUREAU OF STANDARDS

Paper No. 1080	Calibration of Iron Constantan Thermocouple Wire

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washingtom D.C. 20402.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

* 3. REQUIREMENTS

3.1 Qualification. The transmitter furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.6.3 and 6.3).

3.2 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.2.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, and cotter pins, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.2.2 Standard parts. With the exception in 3.2.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.3 Materials.

3.3.1 Materials which are not covered by applicable specifications shall be of the best commercial quality, of the lightest practicable weight, and entirely suitable for the purpose. Nonflammable material shall be used in the construction of the transmitters.

3.3.2 Corrosion resistance. Materials shall be of the corrosion-resisting types or shall be suitably protected to resist corrosion.

3.3.3 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact. Dissimilar metals are defined in MIL-STD-889.

3.3.4 Toxic and corrosive fumes. The materials used in the construction of the transmitter shall not liberate gases which combine with the atmosphere to form acid or corrosive alkali, nor shall they liberate toxic fumes when burned.

3.3.5 Fungus-inert materials. Materials which are not nutrients for fungi shall be used to the greatest extent practicable. Where materials that are nutrients for fungus must be used, such materials shall be treated with a fungicidal agent, as approved by the procuring activity. Materials which are nutrients and contained within a hermetically sealed enclosure need not be treated with a fungicide.

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3.3.6 Protective treatment. When materials are used in the construction of the transmitters that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of atmospheric conditions shall be avoided.

3.3.7 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

3.4 Design and construction.

3.4.1 The transmitter shall conform to the requirements of MS24482 and shall be constructed to withstand the normal strains or jars, vibrations, and such other conditions incident to shipping, storage, installation, and service without failure.

3.4.2 Temperature range. The transmitter shall be designed to function satisfactorily throughout the temperature range -50° to $+300^{o}$ Centigrade (C).

3.5 Performance. The transmitter shall be capable of meeting the test requirements specified in Section 4 when subjected to the following conditions:

a. Temperature. Temperatures ranging from -65° to $+325^{o}$C.

b. Insulation resistance. Satisfactory insulation resistance when a potential of 100 volts (V) direct current (DC) is applied between the connector terminals and any metal part of the transmitter.

c. Vibration. Vibration shall be done in accordance with MIL-STD-810, Section 514.3, Category 4. Fi = 350 Hz

d. Humidity. Relative humidity up to 100 percent.

e. Fungus. Fungus growth as encountered in tropical climate.

f. Salt spray. Exposure to salt sea atmosphere.

g. Spring load. The applied spring load causing the distance between the transmitter tip and the cap-base to be a distance of 2.5 inches for the MS24482-1, and 2.0 inches for the MS24482-2 shall be 2.5 pounds or greater.

3.6 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of DOD-D-1000 shall govern changes in the manufacturer's part numbers.

3.7 Insulation. The wires and insulation outside of the transmitter stem shall be protected by a suitable insulation to resist deterioration from oil, engine cleaning fluids, and heat. The lead wires shall be constructed of flexible stranded wire. A stainless steel flexible sheath or conduit shall be used to protect the leads between the transmitter stem and the metal clamp as indicated on MS24482.

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3.8 Resistance element. The resistance element of the transmitter shall be properly sealed within the transmitter housing to withstand total immersion of the transmitter housing in a liquid bath.

3.9 Temperature-resistance characteristics. The temperature-resistance characteristics of the transmitter shall conform to the values given in Table I.

TABLE I. Temperature-resistance characteristics for cylinder head transmitter.

Temperature °C	Resistance OHMS	Tolerance OHMS	Temperature °C	Resistance OHMS	Tolerance OHMS
-50	37.95	+1.80 -0.60	80	72.65	+1.15 -0.60
-40	40.15	+1.70 -0.60	90	75.80	+1.15 -0.60
-30	42.45	+1.60 -0.60	100	79.00	+1.10 -0.60
-20	44.85	+1.50 -0.60	110 120	82.30 85.70	±0.60 ±0.60
-10	47.35	+1.40 -0.60	130 140	89.20 92.80	±0.60 ±0.60
0	50.00	+1.35 -0.50	150 160	96.50 100.30	+0.70 ±0.70
10	52.60	+1.33 -0.50	170 180	104.20 108.20	±0.70 ±0.70
20	55.25	+1.30 -0.50	190 200	112.30 116.50	±0.70 ±0.80
30	57.95	+1.27 -0.50	210 220	120.80 125.20	±0.80 ±0.80
40	60.70	+1.24 -0.50	230 240	129.70 134.30	±0.80 ±0.80
50	63.55	+1.20 -0.60	250 260	139.00 143.80	±0.90 ±0.90
60	66.50	+1.17 -0.60	270 280	148.75 153.85	±0.90 ±0.90
70	69.55	+1.17 -0.60	290 300	159.10 164.50	±0.90 ±1.00

3.10 Electrical connection. The transmitter shall be provided with electrical connection terminals as specified on MS24482.

3.11 Weight. The weight of the complete transmitter shall not exceed 0.20 pound.

3.12 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The following information shall be included:

MS24482 - (Applicable part number *)

*To be entered by the contractor

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3.13 Workmanship. The transmitter shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, impregnation, and brazing, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsibility for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of tests. The inspection and testing of the transmitter shall be classified as follows:

a. Acceptance tests. Acceptance tests are those tests accomplished on transmitters submitted for acceptance under contract (see 4.3).

b. Qualification tests. Qualification tests are those tests accomplished on samples submitted for qualification as a satisfactory product (see 4.5).

4.3 Acceptance tests. Acceptance tests shall consist of:

a. Individual tests.

b. Sampling plans A and B.

c. Inspection of product.

4.3.1 Individual tests. Each transmitter shall be subjected to the following tests as described under 4.5.

a. Examination of product.

b. Spring load.

c. Resistance check.

4.3.2 Sampling plans and tests.

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4.3.2.1 Sampling plan A. Sampling shall be performed in accordance with MIL-STD-105, inspection Level II, AQL 2.5 percent. Sample units shall be subjected to the following tests as described under 4.5:

a. Individual tests.

b. Transmitter resistance test.

c. High temperature exposure.

d. Insulation resistance.

4.3.2.2 Sampling plan B. Sampling shall be in accordance with MIL-STD-105 special inspection level S-1, AQL 6.5 percent. Sample units shall be subjected to the following tests as described under 4.5:

a. Sampling plan A tests.

b. Temperature gradient.

c. Thermal response.

d. Thermoelectric potential.

e. Self-heating.

f. Low temperature exposure.

g. High temperature exposure of lead insulation.

h. Vibration.

i. Humidity.

j. Fungus

k. Salt spray.

4.3.2.2.1 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of the entire lot shall not be made until it is determined that the lot meets all the requirements of the specification.

4.3.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.3.3.1 Test records. Test records must be available to substantiate the accumulated production quantities.

4.4 Test conditions.

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4.4.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.4.2 Test fixture. When specified, test shall be conducted by using a test fixture with an axial temperature gradient and design equivalent to that shown in Figure 1. The temperature gradient of the fixture shown in Figure 1 shall be approximately 37.8°C (100°F) per inch when the temperature at the bottom of the well is 275°C. The temperature at the bottom of the well shall be determined by means of a master thermocouple located in the fixture wall nearest to the bottom of the well, as shown in Figure 1. The temperature gradient of the fixture shall be determined by the additional use of the thermocouple located 1 inch above the master thermocouple.

4.4.3 Thermocouple EMF. Corresponding values of temperature and electromotive force for iron-constantan thermocouple in accordance with Bureau of Standards Research Paper No. 1080 are shown in Table II.

TABLE II. Iron-constantan thermocouple EMF - temperature characteristics 0°C reference junction.

Temp °C	EMF MV	Temp °C	EMF MV	Temp °C	EMF MV	Temp °C	EMF MV
0	0.00	90	4.85	180	9.87	270	14.90
10	0.52	100	5.40	190	10.43	280	15.46
20	1.05	110	5.95	200	10.99	290	16.01
30	1.58	120	6.51	210	11.56	300	16.56
40	2.12	130	7.07	220	12.12	310	17.12
50	2.66	140	7.63	230	12.68	320	17.67
60	3.20	150	8.19	240	13.23	330	18.22
70	3.75	160	8.75	250	13.79	340	18.77
80	4.30	170	9.31	260	14.35	350	19.32

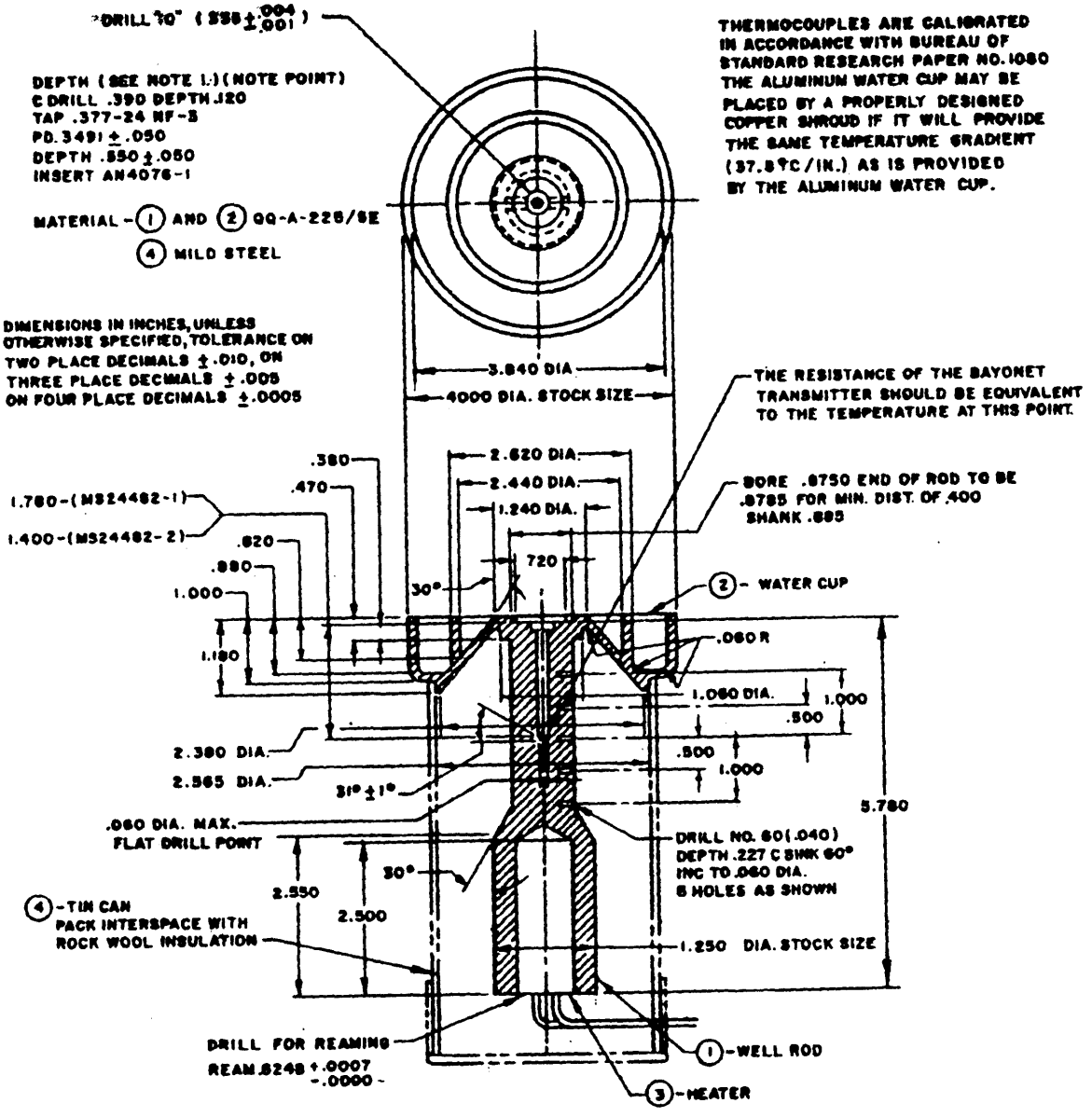
4.4.4 Test voltage. Unless otherwise specified, the transmitter resistance shall be measured with an applied voltage of 1.5 ±0.2V DC.

4.5 Test methods.

4.5.1 Examination of product. Each transmitter shall be inspected to determine compliance with this specification with respect to materials, workmanship, and marking.

4.5.2 Spring load. The tip of the transmitter shall be fixed and the transmitter spring loaded until its length from transmitter tip to case base is 2.50 inches for the MS24482-1 and 2.00 inches for the MS24482-2. The spring load shall be a minimum of 2.50 pounds.

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DEPTH AS SHOWN

BEFORE ASSEMBLY OF (1) -WELL ROD AND (2) -WATER CUP, INSTALL 6 FE-CONST THERMOCOUPLES WITH 2 FT. LEADS AS FOLLOWS: ARC WELD A SHALL BEAD TO FORM JUNCTION ON SPECIALLY SELECTED NO. 30 B & S GAGE, GLASS COVERED, IRON-CONSTANTAN, DUPLEX THERMOCOUPLE WIRE BY ELECTRICAL CONTACT OF TWISTED JUNCTION ON SURFACE OF MERCURY POOL. (USE 12-24V-DC) REMOVE OUTER GLASS BRAID FOR DISTANCE OF .500 FROM TIP, LEAVING INSULATION TO JUNCTION ON INDIVIDUAL WIRES. INSTALL TO FULL DEPTH OF HOLE BY CEMENTING WITH TECHNICAL-B COPPER DENTAL CEMENT. WRAP WIRES $\frac{1}{2}$ TURN AROUND ROD, CEMENT TO ROD SURFACE, WITH GLYPTA-, AND LASH TOGETHER. TAG WIRES TO SHOW LOCATION IN ROD. ASSEMBLE (1) -WELL ROD, AND (2) -WATER CUP, BY PRESSING TOGETHER. INSTALL (3) -HEATER IN BORE, PEEN LIGHTLY TO HOLD IN PLACE BRINGING INTERNAL LEADS THRU, (4) -, NEAR BOTTOM. SPLICE ON 3 FT. OF LAMP CORD AND ATTACH STD 2 PRONG RECEPTACLE PLUG. PACK ROCK WOOL AROUND (1) - AND TIE IN PLACE WITH STRING BEFORE PRESSING ASSEMBLY INTO (4) -COVER.

FIGURE 1. Transmitter test rig.

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4.5.3 Transmitter resistance check. The transmitter shall be immersed to the cap in an agitated liquid bath or other suitable nongradient apparatus at a temperature of 200°C ±0.1°C for a period sufficient to assure that the transmitter is at the test temperature. The test voltage shall be applied across the transmitter terminals and the transmitter resistance measured. The measured resistance shall be 116.50 ±0.89 ohms.

4.5.4 Transmitter resistance test. For acceptance the transmitter shall be immersed to the cap in an agitated liquid bath or other suitable nongradient apparatus at temperature of 0°, 100°, and 300°C for a period sufficient to assure that the transmitter is at the test temperature. The test voltage shall be applied across the transmitter terminals and the transmitter resistance measured. The measured resistance shall be within the tolerances specified in Table I for the test temperature. For qualification, the transmitter shall be immersed at each of the 36 temperatures listed in Table I for sufficient time to assure the transmitter is at the test temperature.

4.5.4.1 When the transmitter resistance test is conducted as part of sampling plan A, tests shall be performed at temperatures of 0°C, 100°C, 200°C, and 300°C (each ±0.1°C).

4.5.4.2 When the transmitter test is conducted as part of sampling plan B, an additional test point of -50° ±0.1°C shall be included.

4.5.5 High temperature exposure. The transmitter shall be inserted in a test fixture with a temperature at the bottom of the well of 325° ±5°C for a period of 8 hours. The temperature at the bottom of the well shall then be reduced to a temperature of 275° ±5°C and the transmitter subjected to this temperature for a period of 42 hours. The transmitter shall then be withdrawn from the fixture and air-cooled to room temperature. The transmitter shall then be subjected to the test specified herein for spring load and the measured spring load shall not be less than 2.00 pounds. The transmitter shall be subjected to the test specified herein for transmitter resistance at a test temperature of 250°C. The measured resistance shall be within the tolerance specified in Table I for the test temperature. The test fixture used for this test need not be as elaborate in design as that shown in Figure 1, but shall have the same axial temperature gradient characteristics. The well of the fixture shall be of minimum depth (7.0 inches).

4.5.6 Insulation resistance. A potential of 100V DC shall be applied for 1 minute between each of the connector terminals and any other metal parts of the transmitter assembly. The insulation resistance shall not be less than 5 megohms.

4.5.7 Temperature gradiant. The transmitter shall be inserted in the test fixture with a temperature at the bottom of the well of 200° ±1°C. Voltage shall be applied across the transmitter and the resistance measured. The measured resistance shall be within 1.7 ohms of the resistance value determined in the transmitter resistance test for the same temperature.

4.5.8 Thermal response. The transmitter shall be inserted in the test fixture with a temperature, at the bottom of the well, of approximately 205°C. The transmitter shall remain in the fixture for a length of time sufficient to assure that the transmitter temperature is approximately the same as the well temperature. The transmitter shall then be removed from the fixture and air

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cooled until the transmitter resistance for 200°C, as determined in the transmitter resistance test is reached. The transmitter shall then immediately be inserted in a second test fixture which has a temperature, at the bottom of the well, of 260° ±1°C. During the time required to properly insert the transmitter in the second fixture, the transmitter temperature will drop slightly below 200°C and therefore the transmitter temperature in the fixture shall be permitted to rise to 200°C before response time is noted. The time required for the transmitter temperature to rise from 200°C to 250°C shall be noted. The resistance values used for transmitter temperature determination at 200°C and 250°C shall be those values determined in the test fixture with temperatures, at the bottom of the well, of 200°C and 250°C. The thermal response of the transmitter shall be considered as an average of three or more test runs and shall not exceed 20 seconds.

4.5.9 Thermoelectric potential. The transmitter shall be inserted in the test fixture with a temperature at the bottom of the well of 200° ±5°C. The potentials produced in the transmitter winding shall not exceed 1 millivolt (mv).

4.5.10 Self-heating. The transmitter shall be inserted in the fixture with a temperature at the bottom of the well of room temperature (approximately 25°C). A current of 50 milliamperes (ma) shall be applied to the transmitter and the rise in resistance measured. The rise in resistance shall not exceed 3.0 ohms.

4.5.11 Low temperature exposure. The complete transmitter assembly shall be subjected to a temperature of -65° ±2°C for a period of 48 hours. A continuity test shall be performed while at this low temperature. No open or short circuit shall occur and no other damage to the transmitter shall result from this test.

4.5.12 High temperature exposure of lead insulation. The entire transmitter assembly shall be subjected to a temperature of 175° ±2°C for a period of 8 hours. There shall be no failure of the lead insulation, nor any other damage to the transmitter assembly.

4.5.13 Vibration. Vibration test to be performed in accordance with MIL-STD-810, method 514.3. The transmitter shall be inserted in a test fixture with a temperature at the bottom of the well of 275° ±10°C and the leads of the transmitter held by a stationary bracket in such a manner as to prevent pull on the leads during vibration. The leads shall be secured in such a manner that the angle between the longitudinal axis of the transmitter and the transmitter leads is a minimum of 100 degrees. The test fixture used for this test need not be as elaborate in design as that shown in Figure 1, but shall be designed to approximate the axial temperature gradient. The well of the fixture shall be of the minimum depth (7.0 inches). Following the vibration test, the following tests, as specified herein, shall be conducted and the results shall be within the tolerance specified for the particular test:

a. Spring load. The spring load shall be a minimum of 2.0 pounds.

b. Transmitter adjustment.

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4.5.14 Humidity. The transmitter shall be subjected to humidity test, Method 507.2, of MIL-STD-810, except for a period of 5 cycles. There shall be no evidence of corrosion or other damage of the transmitter. Following the foregoing humidity test, the test specified herein for insulation resistance shall be conducted and the insulation resistance shall not be less than 1 megohm. Air drying of the transmitter assembly prior to conduction of the insulation resistance test is permissible.

4.5.15 Fungus. The transmitter shall be subjected to a fungus test in accordance with Method 508.3, Procedure 1 of MIL-STD-810. There shall be no evidence of fungus growth on the transmitter as a result of this test and the transmitter shall be subjected to the test specified herein for insulation resistance and the insulation resistance shall not be less than 1 megohm.

4.5.16 Salt spray. The complete transmitter assembly shall be subjected to a salt spray in accordance with Method 509.2 of MIL-STD-810 for a period of 50 hours. At the completion of this test, the transmitter may be cleaned externally with distilled water and air dried. The transmitter shall then be subjected to the tests specified for insulation resistance and transmitter adjustments. The insulation resistance shall not be less than 1 megohm and the transmitter resistance shall not exceed the tolerance specified herein for transmitter adjustment. There shall be no evidence of corrosion or other damage as a result of this test.

4.6 Qualification testing.

4.6.1 Qualification test samples. The qualification test samples shall consist of three transmitters. Samples shall be identified with the manufacturer's own part number and any additional information required by the letter of authorization.

4.6.2 Qualification required. Prior to actual procurement, the product which this specification covers shall pass the qualification tests specified herein. If the product is later modified in any way, the modified form shall be subjected to and shall pass the same qualification tests.

4.6.3 Qualification tests. The qualification tests shall consist of the tests specified under 4.5.

4.7 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of paragraphs 5.1, 5.2, and 5.3 of this specification.

* 5. PACKAGING

* 5.1 Preservation. Preservation shall be level A, C, or Commercial as specified (see 6.2).

* 5.1.1 Level A.

* 5.1.1.1 Cleaning. Transmitter shall be cleaned in accordance with MIL-P-116, Process C-1.

* 5.1.1.2 Drying. Transmitter shall be dried in accordance with MIL-P-116.

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* 5.1.1.3 Preservation applications. Preservatives shall not be used.

* 5.1.1.4 Unit packs. Unless otherwise specified (see 6.2), each transmitter shall be individually unit packed in accordance with method IC of MIL-P-116 designated herein insuring compliance with the applicable requirements of that specification.

* 5.1.1.5 Intermediate packs. Transmitter, unit packed as specified in 5.1.1.4, shall be placed in intermediate containers conforming to PPP-B-566, PPP-B-585, or PPP-B-676. Intermediate containers shall be uniform in size, shape and quantities shall be of minimum tare and cube and shall contain multiples of five unit packs not to exceed 100 unit packs. Intermediate containers shall be closed as specified in the appendices or notes of the applicable container specification. No intermediate packs are required when the total quantity shipped to a single destination is less than 100 unit packs.

* 5.1.2 Level C. The level C preservation for the transmitter shall conform to the MIL-STD-794 requirements for this level.

* 5.1.3 Commercial. The commercial preservation of transmitter shall be in accordance with the requirements of ASTM D 3951.

* 5.2 Packing. Packing shall be level A, B, C or commercial, as specified (see 6.2).

* 5.2.1 Level A. The packaged transmitter shall be packed in containers conforming to PPP-B-601, overseas type: PPP-B-621, class 2, style 4 or PPP-B-585, class 3, style 2 or 3. Closure and strapping shall be in accordance with applicable container specifications except the strapping shall conform to QQ-S-781, type I or IV, finish A. When the gross weight exceeds 200 pounds or the container length and width is 48" X 24" or more and the weight exceeds 100 pounds, 3 X 4 inch skids (laid flat) shall be applied in accordance with the requirements of the container specification.

* 5.2.2 Level B. The packaged transmitter shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirement in the appendix of PPP-B-636, closure waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joints with Waterproof Packaging Tape or Box Closure and Sealing Tape, two inches minimum width.

* 5.2.3 Level C. The level C packing for transmitter shall conform to the MIL-STD-794 requirements for this level.

* 5.2.4 Commercial. The preserved transmitter shall be packed in accordance with the requirements of ASTM D 3951.

* 5.2.5 Palletized loads. When specified (see 6.2) palletized loads, commensurate with the level of packing specified in the contract or order, shall be palletized in accordance with MIL-STD-147. Palletized loads shall be uniform in size and quantities to the greatest extent possible. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

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* 5.3 Marking.

* 5.3.1 Levels A, B, and C. In addition to any special or other identification marking required by the contract (see 6.2), each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

* 5.3.2 Commercial. Commercial markings shall be in accordance with the requirements of ASTM D 3951.

* 5.4 General.

* 5.4.1 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, 5.2.3 and 5.2.4) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.7.

6. NOTES

6.1 Intended use. The transmitters covered by this specification are intended for use in aircraft to indicate, in conjunction with an indicator, the cylinder head temperature of air cooled engine cylinders.

* 6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Whether sampling plan B tests are to be omitted.

* c. Selection of applicable levels of preservation, packaging, packing, marking and commercial/industrial (see 5.1, 5.2, and 5.3).

d. When palletized loads are required (see 5.4).

e. MS 24482-1 or MS 24482-2

6.3 With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List (QPL No.) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is San Antonio ALC/MMIRAC, Kelly AFB TX 78241 and information pertaining to qualification of products may be obtained from that activity.

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6.4 Keyword list.

a. Element, Resistance
b. Temperature, Measuring
c. Temperature, Transmitter
d. Transmitter, Temperature, Cylinder Head
e. Transmitter, Temperature, Electrical Res.
f. Transmitter, Temperature, Flexible Sheath

6.5 Changes from previous issues. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.6 Continuous production. Essentially, continuous production is defined as production of like or similar units with no disruptions of production longer than 90 days. Like or similar units are defined as being of identical basic construction, but differing only in areas such as specific configuration requirements as a type of electrical connector, et cetera.

Custodian:
Air Force - 99

Preparing activity:
Air Force - 82

(Project 6685-F766)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-25446D	Transmitter, Temperature, Electrical Resistance

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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